

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `68286237-c09a-4540-99b6-e253f7edc9e7`
- **Timestamp:** 2026-09-05T16:23:22.701910Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** Which judgments in this corpus apply or construe section 353?
- **Answer:** The retrieved corpus does **not** contain any judicial decisions or case law that apply or construe **section 353** of the Bharatiya Nyaya Sanhita, 2023.

All of the provided chunks are statutory provisions, illustrations, or procedural rules (e.g., sections 3536, 351353, 216217, and the offence table). None of them cite or discuss a court judgment that interprets or applies section 353.

Therefore, there are **no** judgments in this corpus that apply or construe section 353.

Sources: Chunks 0cf5822265bd4215b5109634a4f015e7, 246f091bc03e481c8ee0675521393c74, 2a6630c4f23546ccb83502ca423067a3, d68775e4e16c4412adf4b64a8905d0b1, 6ef00991188c43ee8aa426c06065d78a, 66d3d89000d44f0db42a60e82cf3caf2 (all contain only statutory text, not case law).

- **Health Score:** 0.74
- **Primary Issue:** UNSUPPORTED_GENERATION
- **Claims Supported:** 3 / 6
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
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CORPUS	PASS	0.9	Corpus contains sufficient relevant content, evidenced by at least one supported claim.
RETRIEVER	PASS	0.95	At least one retrieved chunk provides supporting evidence for a verified claim.
GENERATOR	FAIL	0.85	2 generated claims remain unsupported despite high retrieval scores.
GROUNDING	PASS	0.9	No claims are contradicted by retrieved evidence.

Efficiency: 2/6 chunks used (33%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 6 (Supported: 3, Partial: 1, Unsupported: 2, Contradicted: 0)
- **Grounding Score:** 0.58
- **Evidence Coverage:** 0.33
- **Average Entailment:** 0.63
- **Latencies:** Retrieval (2403.4ms) | Generation (47962.6ms) | Verification (301920.0ms)

5. RAGAS Metrics

- **Faithfulness:** 0.5000
- **Answer Relevancy:** 0.8501
- **Context Precision:** 1.0000
- **Context Relevancy:** 0.1667
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[40ecd86-0f62-44d6-9873-250e79c8b47d]** (SUPPORTED): The retrieved corpus does not contain any judicial decisions or 353 of the Bharatiya Nyaya Sanhita, 2023.

- *Evidence (Chunk d68775e4-e16c-4412-adf4-b64a8905d0b1):* 1]

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- **[2a55d864-a97c-4e22-85d3-070de2543b9d]** (UNSUPPORTED): All of the provided chunks are statutory provisions, illustrating 353, 351, 353, 216, 217, and the offence table).

- *Evidence (Chunk 0cf58222-65bd-4215-b510-9634a4f015e7):* 1]

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- **[c3ad8e96-4493-4ffd-aa2a-7a29ec4ee985]** (SUPPORTED): None of the provided chunks cite or discuss a court judgment that mentions 353.

- *Evidence (Chunk d68775e4-e16c-4412-adf4-b64a8905d0b1):* (2) No Court shall take cognizance of—
(a) any offence punishable under section 197 or sub-section (2) or sub-section (3) of section 353 of the Bharatiya Nyaya Sanhita, 2023; or
(b) a criminal conspiracy to commit such offence, except with the previous sanction of the Central Government or of the State Government or of the District Magistrate.

- **[b4d9b1af-5c96-48bd-bc55-292affd75ab2]** (UNSUPPORTED): There are no judgments in this corpus that apply or construe section 353.

- *Evidence (Chunk d68775e4-e16c-4412-adf4-b64a8905d0b1):* 1]

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- **[e1e0b85e-491a-4ffb-a1c2-10fd9e2aee0f]** (SUPPORTED): The sources are chunks 0cf58222, 65bd, 4215, b510, 9634a4f015e7, 246f091b, c03e, 481c, 8ee0, 675521393c74, 2a6630c4, f235, 46cc, b835, 02ca423067a3, d68775e4, e16c, 4412, adf4, b64a8905d0b1, 6ef00991, 188c, 43ee, 8aa4, 26c06065d78a, 66d3d890, 00d4, 4f0d, b42a, 60e82cf3caf2.

- *Evidence (Chunk 0cf58222-65bd-4215-b510-9634a4f015e7):* 1]

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- **[93aeecb7-5472-4fde-b325-3cecf430df6]** (PARTIALLY_SUPPORTED): All of those chunks contain only statutory text, not case law.

- *Evidence (Chunk 0cf58222-65bd-4215-b510-9634a4f015e7):* 1]

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7. Root Cause Analysis

- **Primary Cause:** UNSUPPORTED_GENERATION
- **Confidence:** 0.9
- **Reasoning:**
 - Starting root cause analysis for trace 68286237-c09a-4540-99b6-e253f7edc9e7.
 - CORPUS passed (confidence=0.90).
 - RETRIEVER passed (confidence=0.95).
 - Skipping CHUNKING: no observable state in matrix.
 - Failure at GENERATOR: UNSUPPORTED_GENERATION (confidence=0.85). Observation: 2 generated claims remain unsupported despite evidence.
 - GROUNDING passed (confidence=0.90).
 - Primary root cause: UNSUPPORTED_GENERATION at GENERATOR (earliest upstream failure, confidence=0.85).
 - Analysis complete. Primary cause: UNSUPPORTED_GENERATION.

8. Corrective Action Plan

- **[IMMEDIATE] Lower Sampling Temperature**
 - Description: Reduce generator temperature to 0.0 or 0.1 for deterministic outputs.
 - Expected Improvement: Output becomes rigid and closely aligned to the prompt context. (Metric: Hallucination rate (Unsupported Claims) ↓)
- **[SHORT_TERM] Strict Prompt Grounding**
 - Description: Inject systemic instructions forcing the LLM to output 'I don't know' if context is insufficient.
 - Expected Improvement: LLM reliably declines to answer ungrounded queries. (Metric: Increase in 'Not Verifiable' traces ↓)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
 - Description: Retrieval efficiency is low (33%); consider reducing top-k or tightening the reranker score cutoff.
 - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization ↑)

9. Evidence Traceability

- **Primary Cause:** UNSUPPORTED_GENERATION
- **Derived From:** RootCauseAnalysis
- **Pipeline Stage:** UNSUPPORTED_GENERATION
- **Supporting Claims:** 40ecdc86-0f62-44d6-9873-250e79c8b47d, 2a55d864-a97c-4e22-85d3-070de2543b9d, c3ad8e96-4493-4ffd-aa2b-400000000000
- **Supporting Chunks:** d68775e4-e16c-4412-adf4-b64a8905d0b1, 0cf58222-65bd-4215-b510-9634a4f015e7, d68775e4-e16c-4412-adf4-b64a8905d0b1

10. Overall Assessment

- **Major Strength:** RETRIEVER is healthy (confidence=95%): At least one retrieved chunk provides supporting evidence for the claim.
- **Major Weakness:** UNSUPPORTED_GENERATION
- **Next Priority:** Lower Sampling Temperature
- **Recommendation:** Pipeline requires targeted improvements to the primary failure stage.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework
- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ ClaimSet

artifacts/claim_sets/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ VerificationResults

artifacts/verification/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/68286237-c09a-4540-99b6-e253f7edc9e7.json

✓ DiagnosticEvaluationReport

artifacts/reports/68286237-c09a-4540-99b6-e253f7edc9e7.json